Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/10/2025
PI Name:	Allaway, Heather Ph.D.		
Project Title:	Insights Into the Impacts of Continuous, Low Dose-Rate Neutron Radiation Exposure on Maternal and Fetal Skeletal Physiology		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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City:	Baton Rouge	**State:**	LA
Zip Code:	70803-0001	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	06/01/2023	**End Date:**	05/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/31/2026 per F. Hernandez/ARC (Ed., 2/21/26). NOTE: End date changed to 05/31/2025 per NSSC information (Ed., 5/22/24).		
Key Personnel Changes/Previous PI:	No Key Personnel Changes were made during the reporting period		
COI Name (Institution):	Lau, Anthony Ph.D. (College of New Jersey)		
Grant/Contract No.:	80NSSC23K0784		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The combined effects of space environmental stressors induce pathologies in multiple organ systems. Specifically, losses to the musculoskeletal system may be very dangerous for the health and performance of astronauts on extended duration missions on the Moon or arriving on Mars. A critical need remains to better understand the impact of radiation exposure, one of the key environmental stressors of deep space, on human and animal physiology to enable extended duration missions beyond low Earth orbit or setting up settlements on the Moon or Mars. There is a critical gap in knowledge surrounding the impact of the space radiation environment on skeletal health and on the progress of fetal skeletal development during pregnancy. The objective of the current proposal is to capitalize on a unique tissue-sharing opportunity to examine the combined effects of continuous radiation exposure and pregnancy on maternal and fetal skeletal physiology. We propose to assess maternal and fetal skeletal physiology through measurements of mineral and material properties, as well as assess changes in cellular dynamics of the maternal bone under the microscope. This study will be critical in assessing how a very harmful component of the space radiation environment impacts multiple aspects of skeletal health, including sex-specific differences and individual variation in the impact of the space environment on the functioning of the body.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides new insights on how near continuous low-dose of ionizing radiation (<0.1Gy) affects the maternal fetal system related to bone. The maternal and fetal skeletal changes from these extremely low doses could be considered exposure to clinical radiation for diagnostic and radiotherapies.
Task Progress:	During the past reporting period, the assessment of fetal sample external morphology (AIM 2) continued, dissecting microscope images obtained, and microCT scanning (E18 samples only) began. Processing of the microCT scans has just begun. We continued the search for a research lab or company to perform the sex determination genetic testing of the samples and were finally able to locate a reputable commercial lab. We purposefully delayed all types of destructive analysis until all possible non-destructive analyses were complete. All available fetal data was presented at local conferences and the NASA Human Research Program (HRP) Investigators' Workshop (IWS). MicroCT scanning of the maternal tibia (AIM 1) has been completed at the proximal metaphyseal area and midshaft. Our team is actively working on analyses of the scans. All tibia and femur samples were shipped to the collaborator. 3-point-bend to failure mechanical testing has been completed and microindentation testing is underway. Plastic and paraffin embedding of the samples and the re-embedding is complete. Training of a new student is underway to complete the histologic sectioning; this process is slow as the samples are very rare. Issues with the computer attached to the microscope has delayed training for anaysis of all histologic sections.
Bibliography Type:	Description: (Last Updated: 01/28/2026)
Abstracts for Journals and Proceedings	Sorrels C, Steller J, LoGiudice M, Lau AG, Chancellor J, Almeida E, Allaway H. "Insights into the impacts of continuous, low-dose-rate neutron radiation exposure on fetal skeletal development." LaSPACE Fall 2024 Annual Conference, Louisiana Space Grant Consortium, Hammond, LA, October 5, 2024. Abstracts. LaSPACE Fall 2024 Annual Conference, Louisiana Space Grant Consortium, Hammond, LA, October 5, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Liegner JK, Bowman LC, Lau AG, Allaway HCM. "Effects of continuous, low dose-rate neutron radiation exposure during pregnancy on maternal bone microstructure." Biomedical Engineering Society Annual Meeting, Baltimore, MD, October 23-26, 2024. Abstracts. Biomedical Engineering Society Annual Meeting, Baltimore, MD, October 23-26, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Sorrels C, Steller J, LoGiudice M, Lau AG, Chancellor J, Almeida E, Allaway HCM. "Insights into the impacts of continuous, low-dose-rate neutron radiation exposure on fetal skeletal development." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Liegner JK, Bowman LC, Lau AG, Allaway HCM. "Effects of continuous, low dose-rate neutron radiation exposure during pregnancy on maternal bone microstructure." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Sorrels C, Steller J, LoGiudice M, Lau AG, Chancellor J, Almeida E, Allaway HCM. "Impacts of continuous, low-dose-rate neutron radiation exposure during gestation on fetal development." National Conference on Undergraduate Research, Pittsburgh, PA, April 7-9, 2025. Abstracts. National Conference on Undergraduate Research, Pittsburgh, PA, April 7-9, 2025. , Apr-2025
Awards	Sorrels, C. "Travel Stipend, Louisiana State University (LSU) Office of Undergraduate Research." Apr-2025
Awards	Sorrels, C. "LaSPACE Undergraduate Research Assistantship (LURA), August 2024." Aug-2024